$\boxed{\text{Solution:}}$ The following 4 cases arise:

$\boxed{\text{Case 1:}}$ $x \geq 0, y \geq 0$. $\therefore |x| = x, |y| = y$. (By defn. of absolute value)

$\boxed{\therefore xy \geq 0}$ $|xy| = xy$ (By defn. of absolute value)

$= |x| \cdot |y|$

$\boxed{\text{Case 2:}}$ $x \geq 0, y < 0$ $\therefore |x| = x, |y| = -y$ (By defn. of absolute value)

$\therefore xy < 0$ $|xy| = -(xy)$ (By defn. of absolute value)

$= x \cdot (-y) = |x| \cdot |y|$

$\boxed{\text{Case 3:}}$ $x < 0, y \geq 0$ $\therefore |x| = -x, |y| = y$ (By defn. of absolute value)

$\therefore xy < 0$ $|xy| = -(xy)$ (By defn. of absolute value)

$= (-x) \cdot y = |x| \cdot |y|$

$\boxed{\text{Case 4:}}$ $x < 0, y < 0$ $\therefore |x| = -x, |y| = -y$ (By defn. of absolute value)

$\boxed{\therefore xy > 0}$ $|xy| = xy = (-x) \cdot (-y) = |x| \cdot |y|$

(By defn. of absolute value)

Because $|xy| = |x| \cdot |y|$ holds in each of the four cases and these cases exhaust all possibilities, we can conclude that $|xy| = |x||y|$, whenever $x$ and $y$ are real numbers. $\square$

$\boxed{\text{Leveraging Proof by Cases}}$ The examples we have presented illustrating proof by cases provide some insight into when to use this method of proof. In particular, when it is not possible to consider all cases of a proof at the same time, a proof by cases should be considered.

Q) When should we use such a proof?